

Salmon Spawning Behavior Analysis using MCMC

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1 Introduction

The project aims to investigate life-cycles of salmon at the pacific northwest region, and provide useful insights to fisheries management. Although there are couple of ways to analyze salmon life cycle, main focus of this particular research is modelling the spawning behaviors of salmon. For this purpose, one might need to know the type of a salmon (odd-year or even-year) as it is probably the most important predictor of the spawning numbers. One particular issue here is that type information is not available for us in the given data set, thus we have to build model so that it also infers the type, and then we should integrate it into the model which explores the spawning behavior of a salmon. By doing so, we will have an idea about the spawning behavior for a specific type of a salmon, and then we can relate this information to other components of the model. Some of the specific questions that this model aims to answer are as follows.

- How much do the certain factors affect the spawning behavior?
- Are they negatively or positively related to the spawning behavior?
- Is spawning and recruitment related?
- What is the difference between mean spawning numbers of odd-year and even-year salmon?

2 Methods

Before starting with the modelling, one needs to focus on missing and probably misleading parts in data. First of all, part of the missing data is related to the stock name "Snohomish". However, not all "Snohomish" related data is missing. We know that their spawning region is "Puget Sound". Therefore, one can fill the missing values based on the region information since region uniquely determines the latitude, longitude and alongshore distance. Since "Snohomish" is retained in the model, we add one more stock number, which is also missing for "Snohomish", and call it 44. Moreover, stock number 5 has wrong information, as there are two different regions related to stock number even though

it should be unique for each stock number. Since region of "Puget Sound" occurs more frequently for stock number 5, one can consider all of the regions as "Puget Sound" assuming that less frequent ones are not correct.

The recruitment and spawning numbers are key (most informative) observations in the given data set. These two columns captures all the information about the life-cycles of salmons for each region, salmon type and stock number. Therefore, the model in this paper is based on spawning and recruitment numbers. However, although we know that they are particularly important for understanding the salmon behavior, we don't have an idea about how other features help us to understand salmon life-cycle. Hence, a linear model that accounts different effects with different coefficients is designed for this purpose. Before defining the model mathematically, some of the existing relationships in the data set should be noted. Firstly, based on the figure 1, we can say that spawning numbers and recruitment are clearly correlated, which means that we can focus on either. Additionally, including one of them as a predictor make the model unfair because of the correlation. However, when we compare number of spawning with row differences in R (row-wise differences of number of recruitment with a window of 1), we don't observe same relation (See figure 1 (right)). Therefore, we can integrate row-wise differences of recruitment number in the inference of spawning numbers. In fact, this approach helps model to mitigate the temporal dependence caused by time-series effect.

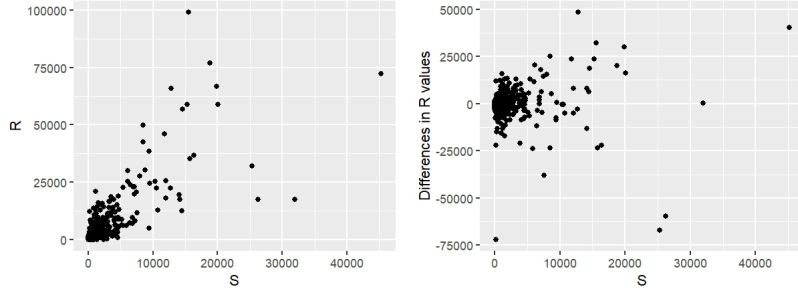


Figure 1: (Left) Number of spawning vs. number of recruitment, (Right) Number of spawning vs. differences in number of recruitment

Another important factor for spawning numbers is the type of the salmon, i.e. if it is odd-year or even-year salmon. However, this information is not available for us, so we need to add extra layer to the hierarchical model to count this effect in the model. One can use following model for this purpose where $type = 1$ means even-year salmon and $type = 0$ means odd-year salmon. This model is valid for this particular variable as type is modelled as binary, so Bernoulli distribution is correct choice. Additionally, again, we don't know what actually affects the probability of being even or odd year salmon, and we can overcome this issue by using logistic regression approach and mapping the probability measure to the whole real line with logit function and constructing

another linear model for $\text{logit}(\theta)$ as in equation 1.

$$\begin{aligned} \text{type}_{i,j} &\sim \text{Bernoulli}(\theta_{i,j}), \\ \text{logit}(\theta_{i,j}) &= \alpha X. \\ \text{where } i &\in \{1, \dots, N_{\text{stock}}\} (\text{stock numbers}), \quad j \in \{1, \dots, n_i\} \end{aligned} \quad (1)$$

One possible way to effectively model the probability of being an even or odd year salmon using logit link function can be seen in equation 2.

$$\begin{aligned} \text{logit}(\theta_{i,j}) &= \alpha_0 + \alpha_1 R2_{i,j} + \alpha_2 (\text{odd-even}), \\ \text{where } R2_{i,j} &: \text{differences in R but with a window of 2 rows} \\ \text{where } i &\in \{3, \dots, 44\} (\text{stock numbers}), \quad j \in \{1, \dots, n_i\} \end{aligned} \quad (2)$$

Note that row differences of R are again taken into account as the type information is roughly based on change in the number of spawning or recruitment. However, this time 2 year differences are considered. As we discussed above, one year differences will be incorporated into the model for spawning numbers. If we also considered same variable here, we would obtain correlation issue at the end. However, 2 year differences and 1 year differences seem to be unrelated (See figure 2 in Appendix).

One another model component is an engineered variable which is called "competition", which can be described as

$$\text{competition} = \frac{\# \text{ of different stock numbers in a region}}{\text{Alongshore distance in a region}}$$

This particular measure indicates how many different salmon species compete for recruiting, so we expect a negative relation between competition and spawning number. In fact, what we observe in figure 3 in Appendix agrees with this expectation. Note that competition is actually indexed with region, but since each stock number is related to only one region we can index it with i (stock numbers) to make the model more interpretable.

All in all, complete model is described as follows.

$$\begin{aligned} p(S|\beta_0, \beta_1, \beta_2, \beta_3, \sigma^2) &\sim \prod_{i=1}^{N_{\text{stock}}} \prod_{j=1}^{n_i} \mathcal{N}(S_{i,j}; \mu_{i,j}, \sigma_i^2) \\ \mu_{i,j} &= \beta_0 + \beta_1 (\text{type}_{i,j}) + \beta_2 (\text{competition}_i) + \beta_3 (\text{R-differences})_{i,j} \\ \text{type}_{i,j} &\sim \text{Bernoulli}(\theta_{i,j}), \quad \text{logit}(\theta_{i,j}) = \alpha_0 + \alpha_1 R2_{i,j} + \alpha_3 (\text{odd-even})_{i,j} \\ \beta_0, \beta_1, \beta_3 &\stackrel{\text{iid}}{\sim} \mathcal{N}(0, 1000), \quad \beta_2 \sim \mathcal{N}(-10, 1000), \quad \sigma^2 \sim \prod_{i=1}^{N_{\text{stock}}} \text{Inv-Gamma}(\sigma_i^2; 100, 100) \\ \alpha_0, \alpha_1, \alpha_2 &\stackrel{\text{iid}}{\sim} \mathcal{N}(0, 1000) \end{aligned}$$

In this model, we particularly consider $\beta_i, \alpha_j \sim \mathcal{N}(\cdot, \cdot)$, $\forall i \in \{0, \dots, 3\}$, $\forall j \in \{0, \dots, 2\}$. Main reason for using normal distribution for regression coefficients is the fact that they are defined on $\mathbb{R}$. Note that parameters for these distributions are selected so that these distribution are vague as much as possible. However, since we have a prior information about the relationship between competition and number of spawning, we integrate this knowledge by taking the mean of normal distribution as negative. Additionally, σ^2 is considered to be distributed with Inverse-Gamma since it is a conjugate prior for the variance of normal distribution. However, we again consider weakly informative distribution as there is some available prior knowledge. We know that S is in thousands (it's quite big in magnitude), therefore we expect that its variance should also be in a large scale, so we choose parameters of this prior as 100.

In the model, we consider "competition" as fixed, not random. However, one can also model it with Dirichlet distribution and then integrate it to the mean function. However, after some trials, extra Dirichlet layer caused problems for the convergence and requires huge number of iterations to truly obtain convergence which is not quite feasible. See appendix section 5.4, for the resulting trace plot in that case, which indicates that algorithm definitely needs more run for convergence.

3 Results

We need to first investigate if given MCMC algorithm converges to the desired posterior distribution. For this purpose, one can check the diagnostic plots (See Appendix section 5.2). When the model is compiled with total iteration number of 30000 and burn-in period of first 7500 iterations, we observe the following trace plots (More iteration would probably result in better representing posterior samples but current computing equipment doesn't allow more runs and R-markdown faces inevitable fatal errors in that case). These plots don't indicate any particular issue with the parameters, and convergence is obtained. Similarly, density plots are not concerning for the convergence properties of the chain. Note that, some of the variables in the model (σ^2, θ etc.) have many levels, and plotting each of them separately takes so much space and makes the interpretation complicated. Therefore, we choose some of the levels randomly and plot them by assuming that rest of them will also be similar in a sense.

4 Discussion

4.1 Answers for the questions

- **Question 1: How much do the certain factors affect the spawning**
We can answer this question by estimating the coefficients in the regression equation for mean of spawning numbers. Since stock numbers and the measurements for each stock numbers combined make it hard to interpret, we can construct this equation for specific stock number. The

fisheries management can easily modify the input and get the estimations for desired stock numbers. Considering the stock number 4, model estimates (mean of type(4,)) that it is type 0, which means it is an odd-year salmon. When we plug the mean of β values, mean function becomes

$$E(\bar{S}_{4,.}|\text{stock}) = 30.14 - 16.55 * \text{competition}_4 + 0.03 * \text{Rdifferences}_{4,.}$$

One can see that model considers recruitment differences (with a window of 1) as positively related to the number of spawning. One should be aware of the fact these estimations are not fixed, and change with each run.

- **Question 2: Are the type estimations reasonable?:** The figures and table in appendix section 5.3 indicates that type estimations are fairly reasonable for some selected stock numbers. For instance, for stock number 39, model says it is type 1, and based on the plot we observe that it is true. Similarly for stock number 27, although spawning numbers are sometimes higher for odd years, we generally obtain increase in S for even years, which agrees with model estimation. Note that model sometimes fails to estimate type (e.g. for stock number 29). However, we mostly observe that model correctly estimates the type.
- **Question 4: What is the difference between mean spawning numbers of odd-year and even-year salmons?** When, we take the average of estimated mean of spawning number over each salmon based on their type we obtain that mean spawning number of odd-year salmon is 86.27, and for even year it is 152.5. Therefore, we expect more spawnings in the even years based on the model outputs. The calculation is like following.

$$\text{mean spawning numbers for type 1} = \frac{1}{M} \sum_{k=1}^M \mu_{i,.}^{(k)} \mathbb{I}(\text{type} = 1)$$

$$\text{mean spawning numbers for type 0} = \frac{1}{M} \sum_{k=1}^M \mu_{i,.}^{(k)} \mathbb{I}(\text{type} = 0)$$

4.2 Limitations and discussions

There are couple of disadvantages of this study. One them is related to recruitment differences. When, we take differences by row, the last row of R -differences for each stock number is actually wrong because we simply take difference for two separate stock numbers. This might be ignorable given we have enough observations for each stock number, but when this is not the case it should be carefully adjusted. Another disadvantage is that, we consider more than one variable in the model as fixed (R differences, competition etc.). However, assigning prior to them and allowing chain to explore more in the parameter space would be a better choice. However, given the computational constraints, this simpler version fits better for our purposes. One further research direction might be utilizing this model for prediction, which probably

would be more useful for fisheries management. This can be done with small adjustments in the code.

5 Appendix

5.1 Data related plots

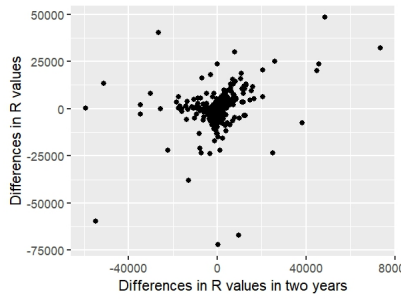


Figure 2: 2 years differences in R vs. 1 year differences in R

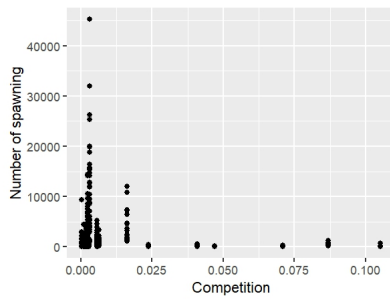


Figure 3: Competition vs. Number of spawning

5.2 Model diagnostics

5.2.1 Trace plots

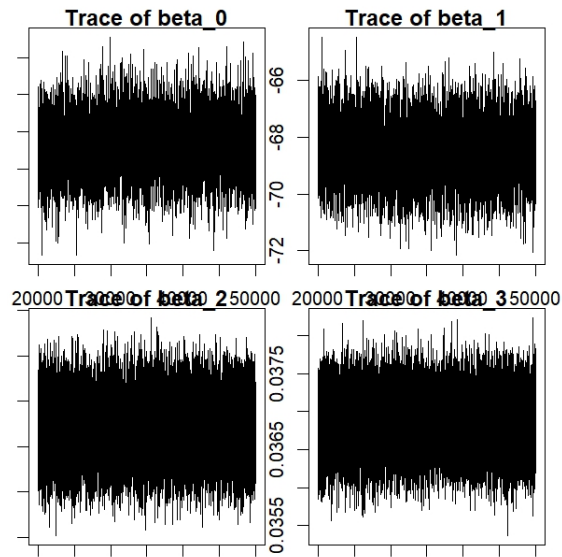


Figure 4: Trace plots for β variables

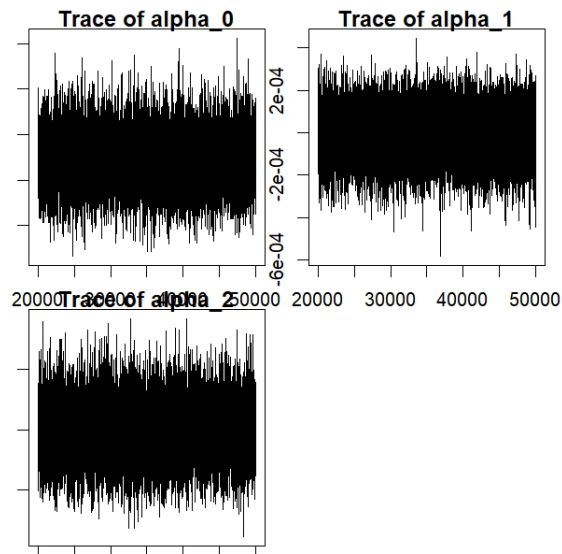


Figure 5: Trace plots for α variables

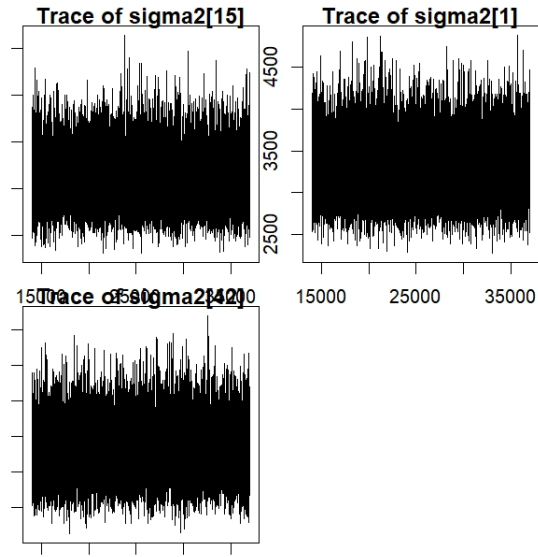


Figure 6: Trace plots for some of the σ^2 variables

5.2.2 Density plots

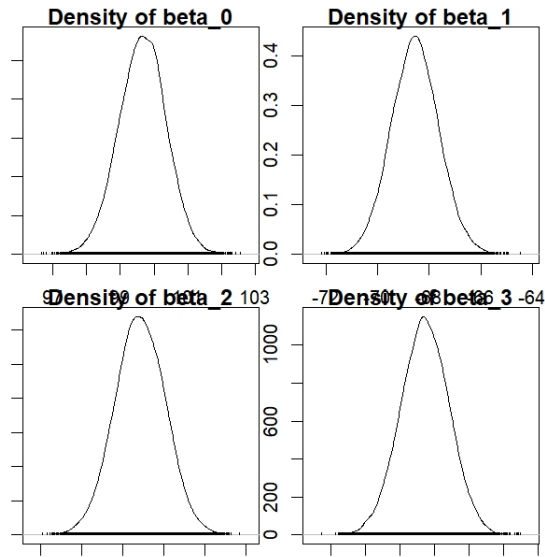


Figure 7: Trace plots for β variables

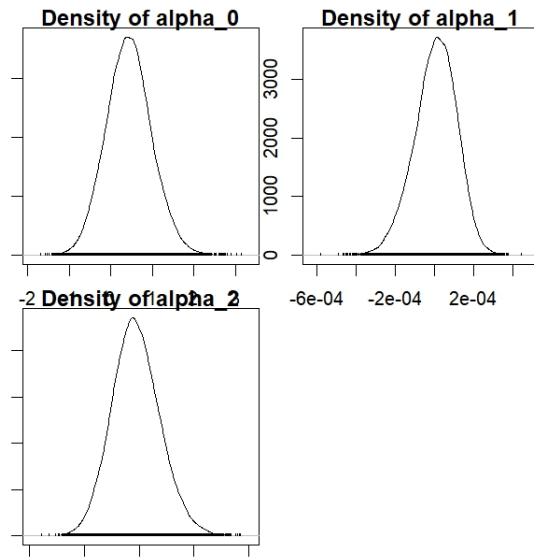


Figure 8: Trace plots for α variables

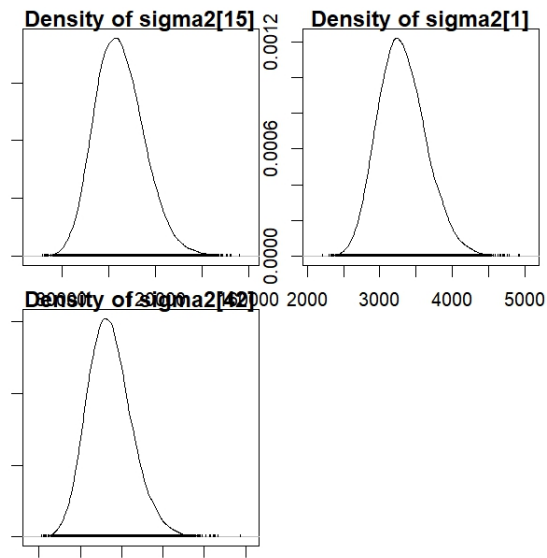


Figure 9: Trace plots for some of the σ^2 variables (indexes: 1,15,42)

5.3 Are type estimations reasonable?

type[39,15]	1.00000
type[24,17]	1.00000
type[25,17]	1.00000
type[26,17]	1.00000
type[27,17]	1.00000
type[28,17]	1.00000
type[37,17]	0.99997
type[38,17]	0.00000
type[1,19]	1.00000
type[2,19]	0.00000
type[4,19]	1.00000
type[5,19]	0.00000
type[42,19]	1.00000
type[6,20]	0.99957
type[7,21]	1.00000
type[17,21]	1.00000
type[41,21]	1.00000
type[40,22]	0.00000
type[16,25]	1.00000
type[19,26]	1.00000
type[18,30]	1.00000
type[20,30]	1.00000
type[29,31]	0.00000
type[23,34]	1.00000
type[30,34]	0.00000
type[31,34]	1.00000

Figure 10: Posterior means for type variables for each stock number



Figure 11: (Left) Spawning numbers of stock number 39 grouped by year type, (Right) same plot for stock number 27.

5.4 What if competition is modelled with Dirichlet?

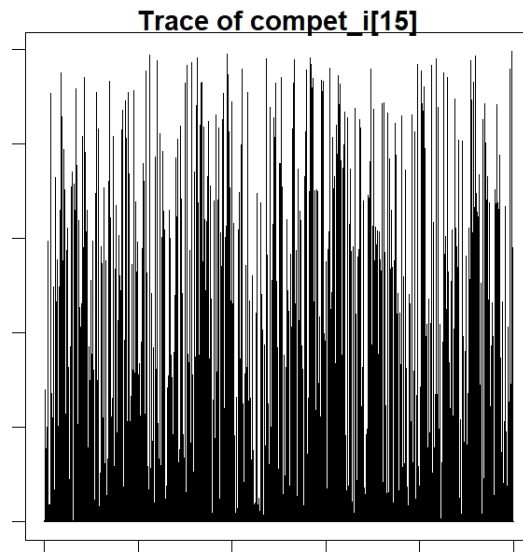


Figure 12: Trace plots for only one pr variables when Dirichlet distribution is used additional layer for modelling the probabilities

5.5 Code

```
library(broom)
library(dplyr)
library(coda)
library(rjags)
library(tidyverse)
par(mar=c(1,1,1,1))

# read in .csv data
data<-read.csv('pinkdata.csv')
# add vector of observation numbers
data[['obs']]<-seq(1,dim(data)[1])
# create a new variable S and give it the values in S
data[['S']]<-data$S
# create a new variable R and give it the values in S
data[['R']]<-data$R
# create a new variable yr and give it the values in BY
data[['yr']]<-data$BY

# convert list to data frame
df.pink<-data.frame(data)
```

```

#Filling NAN values based on region and stock
df.pink[df.pink$Stock=="Snohomish",]$Latitude = mean(df.pink[df.pink$Region=="Puget Sound",]$Latitude)
df.pink[df.pink$Stock=="Snohomish",]$Longitude = mean(df.pink[df.pink$Region=="Puget Sound",]$Longitude)
df.pink[df.pink$Stock=="Snohomish",]$AlongShore_Distance = mean(df.pink[df.pink$Region=="Puget Sound",]$AlongShore_Distance)
df.pink[df.pink$Stock=="Snohomish",]$StockNumber = 44

target_cat <- "Puyallup"
df.pink[df.pink$Stock==target_cat & is.na(df.pink$Latitude),]$Latitude=df.pink[df.pink$Stock==target_cat & is.na(df.pink$Latitude),]$Latitude
df.pink[df.pink$Stock==target_cat & is.na(df.pink$Longitude),]$Longitude=df.pink[df.pink$Stock==target_cat & is.na(df.pink$Longitude),]$Longitude
df.pink[df.pink$Stock==target_cat & is.na(df.pink$StockNumber),]$StockNumber=df.pink[df.pink$Stock==target_cat & is.na(df.pink$StockNumber),]$StockNumber
df.pink[df.pink$Stock==target_cat & is.na(df.pink$AlongShore_Distance),]$AlongShore_Distance=df.pink[df.pink$Stock==target_cat & is.na(df.pink$AlongShore_Distance),]$AlongShore_Distance

# create a new variable asd and give it the values in AlongShore_Distance
data[,'asd']<-data$AlongShore_Distance

# stock number 5 have two different locations set it to most frequent one, which is Puget Sound
df.pink[df.pink$StockNumber==5,]$Region="Puget Sound"
df.pink[df.pink$StockNumber==5,]$Longitude=48.23861
df.pink[df.pink$StockNumber==5,]$Latitude=122.3758
df.pink[df.pink$StockNumber==5,]$AlongShore_Distance=38

# differences
df.pink$S_diffs <- c(df.pink$S[1], diff(df.pink$S))
df.pink$R_diffs <- c(df.pink$R[1], diff(df.pink$R))

# odd even year
df.pink$odd_even = ifelse(df.pink$yr %% 2 == 0, 1,0)

# stock count
temp_df = df.pink %>% group_by(Region) %>% summarize(stock_count = n_distinct(StockNumber))
df.pink = merge(df.pink,temp_df,by="Region")

gcdcalc<-function(lon1,lat1,lon2,lat2)
{
  r<-6378
  gcd<-r*acos(sin(lat1*pi/180)*sin(lat2*pi/180)+cos(lat1*pi/180)*cos(lat2*pi/180)*cos((lon1-lon2)*pi/180))
  gcd
}

# find southernmost stock location
zerogcd<-min(df.pink[, 'Latitude'], na.rm=TRUE)

```

```

smost.ind<-which(df.pink[, 'Latitude']==zerogcd)[1]
# measure Great Circle Distance to all other stock locations from Southernmost stock
df.pink[, 'gcd']<-gcdcalc(df.pink[smost.ind, 'Longitude'], df.pink[smost.ind, 'Latitude'], df.pir

df.pink$S_2_diffs = c(df.pink$S[1], df.pink$S[2], diff(df.pink$S, lag = 2))
df.pink$R_2_diffs = c(df.pink$R[1], df.pink$R[2], diff(df.pink$R, lag = 2))

df.pink$SR_2_diffs = (df.pink$S_2_diffs + df.pink$R_2_diffs)/2

species = seq(1, length(unique(df.pink$StockNumber)))

temp_df <- df.pink %>% group_by(StockNumber) %>% summarize(count=n()) %>% as.data.frame()
ni = temp_df$count

Rdiffs_by_sppecies <- tapply(df.pink$R_diffs, df.pink$StockNumber, FUN = function(x) x)
max_length <- max(sapply(Rdiffs_by_sppecies, length))
Rdiffs_ij <- sapply(Rdiffs_by_sppecies, function(x) {
  length(x) <- max_length
  x
})
Rdiffs_ij <- matrix(unlist(Rdiffs_ij), nrow = length(Rdiffs_by_sppecies), byrow = TRUE)
Rdiffs_ij[is.na(Rdiffs_ij)] = 0

R2diffs_by_sppecies <- tapply(df.pink$R_2_diffs, df.pink$StockNumber, FUN = function(x) x)
max_length <- max(sapply(R2diffs_by_sppecies, length))
R2diffs_ij <- sapply(R2diffs_by_sppecies, function(x) {
  length(x) <- max_length
  x
})
R2diffs_ij <- matrix(unlist(R2diffs_ij), nrow = length(R2diffs_by_sppecies), byrow = TRUE)
R2diffs_ij[is.na(R2diffs_ij)] = 0

odd_even_by_sppecies <- tapply(df.pink$odd_even, df.pink$StockNumber, FUN = function(x) x)
max_length <- max(sapply(odd_even_by_sppecies, length))
odd_even_ij <- sapply(odd_even_by_sppecies, function(x) {
  length(x) <- max_length
  x
})
odd_even_ij <- matrix(unlist(odd_even_ij), nrow = length(odd_even_by_sppecies), byrow = TRUE)
odd_even_ij[is.na(odd_even_ij)] = 0

S_by_sppecies <- tapply(df.pink$S, df.pink$StockNumber, FUN = function(x) x)
max_length <- max(sapply(S_by_sppecies, length))
S_ij <- sapply(S_by_sppecies, function(x) {
  length(x) <- max_length

```

```

      x
    })
    S_ij <- matrix(unlist(S_ij), nrow = length(S_by_species), byrow = TRUE)
    S_ij[is.na(S_ij)] = 0

    df.pink$competition = df.pink$stock_count/df.pink$AlongShore_Distance
    competition_i = tapply(df.pink$competition,df.pink$StockNumber,mean)
    competition_i_probs = competition_i/sum(competition_i)

    # i: stock number, j: observations for each stock number

    # lower competition, higher the chances of survival
    # so higher number of spawning in the consecutive years. add plot of competition vs S

    project_model = "model{
      # likelihood
      for (i in species){
        for (j in ni[i]){
          S_ij[i,j]~dnorm(mu[i,j],tau2[i])
          mu[i,j]=beta_0 + beta_1*type[i,j]+beta_2*competition_i[i]+beta_3*Rdiffs_ij[i,j]
        }
      }

      beta_0~dnorm(0,0.1)
      beta_1~dnorm(0,0.1)
      beta_2~dnorm(-10,0.1)
      beta_3~dnorm(0,0.1)

      for (i in species){
        # since S in thousands (it's quite big in magnitude), variance should also be
        # in a large scale
        tau2[i]~dgamma(100,100)
        sigma2[i] = 1/tau2[i]
      }
      for (i in species){
        for (j in ni[i]){
          type[i,j]~dbern(theta[i,j])
          theta[i,j]=ilogit(z[i,j])
          z[i,j] = alpha_0 + alpha_1*R2_diffs_ij[i,j]+alpha_2*odd_even_ij[i,j]
        }
      }

      alpha_0 ~ dnorm(0,0.001)
      alpha_1 ~ dnorm(0,0.001)
      alpha_2 ~ dnorm(0,0.001)

```

```

    for (i in species){
      for (j in ni[i]){
        indicator[i,j] <- ifelse(type[i,j]>0,0,1)
        mu_est1[i,j] <- (beta_0 + beta_1*type[i,j]+beta_2*competition_i[i]+beta_3*Rdiffs_ij)
        mu_est0[i,j] <- (beta_0 + beta_1*indicator[i,j]+beta_2*competition_i[i]+beta_3*Rdiffs_ij)
      }
    }
  }
}"

model_init= list(beta_0=0,beta_1=0,beta_2=-10,beta_3=0,tau2=rep(2,length(ni)),
                 alpha_0=0, alpha_1=0, alpha_2=0)
nitters=30000
nburns=7000
nadapt=7000

fit_data = list(S_ij=S_ij,species=species,Rdiffs_ij=Rdiffs_ij,R2_diffs_ij=R2_diffs_ij,
                odd_even_ij=odd_even_ij,competition_i=competition_i,ni=ni)

fit = jags.model(textConnection(project_model),data=fit_data,init=model_init,n.chains = 1,n.burns = 7000)
#fit.samples_sigma = coda.samples(fit,c("sigma2[1]","sigma2[15]","sigma2[42]"),n.iter = nitters)
#fit.samples_beta = coda.samples(fit,c("beta_0","beta_1","beta_2","beta_3"),n.iter = nitters)
#fit.samples_alpha = coda.samples(fit,c("alpha_0","alpha_1","alpha_2"),n.iter = nitters)
#fit.samples_theta = coda.samples(fit,c("theta[1,1]","theta[1,10]","theta[42,10]","theta[42,100]"),n.iter = nitters)

fit.samples_estS = coda.samples(fit,c("type","beta_0","beta_1","beta_2","beta_3","mu_est1","mu_est0"),n.iter = nitters)

#summary(fit.samples_sigma)
#summary(fit.samples_beta)
#summary(fit.samples_alpha)

#plot(window(fit.samples_sigma,start=nburns+nadapt),trace = F,density = T)
#plot(window(fit.samples_beta,start=nburns+nadapt),trace = FALSE,density = TRUE)
#plot(window(fit.samples_alpha,start=nburns+nadapt),trace = F,density = T)

```